

Technical Document #970: Natural Language Processing - Comprehensive Overview

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Question answering systems extract answers from a given context. They typically consist of a retriever component and a reader component.

Machine translation converts text from one language to another. Neural machine translation using encoder-decoder architectures has achieved human-level performance.

Sentiment analysis determines the emotional tone behind a body of text. It is widely used in social media monitoring and customer feedback analysis.

Language models predict the probability of a sequence of words. Large language models like GPT and BERT have achieved remarkable performance on various NLP tasks.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Key Takeaways:

- Language models predict the probability of a sequence of words.
- Text summarization generates a concise summary of a longer text.
- Coreference resolution identifies all expressions that refer to the same entity in a text.